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National Institute of Technology Goa

Programme Name: B.Tech
End Semester Examinations, 2024-25

Course Name: **Differential Equations and Vector Calculus**
Date: 06/05/2025
Duration: 3 Hours

Course Code: **MA150**
Time: 10:00 A.M - 01:00 P.M
Max. Marks: 100

1. Answer All Questions.
2. **No marks will be given if the explanation of your answer is missing.**
3. The question paper consists of **two** pages.
4. Write the answers to all the questions of each part in one place.

Part A (Ordinary Differential Equations) [5 × 5 = 25]

1. Consider the differential equation $(x^3y^3 + x^2y^2 + xy + 1)dx + (x^3y^3 - x^2y^2 - xy + 1)dy = 0$.
(a) Check whether the given ODE is exact or not?
(b) If yes find its general solution.
(c) If not find the I.F. and its general solution. [5M]
2. Solve the Bernoulli's differential equation $(x^2 + 2y^2 - 2x)dx + 2xydy = 0$. [5M]
3. Find the solution of Cauchy-Euler equation $x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{(1-x)^2}$. [5M]
4. Using the method of undetermined coefficients find the trial solution of $(D^2 - 4D + 4)y = x^3e^{2x} + xe^{2x}$. [5M]
5. Solve the following system of equations using linear algebra [5M]

$$\begin{aligned}\frac{dx_1}{dt} &= x_1 + x_2, \\ \frac{dx_2}{dt} &= x_2 + x_3, \\ \frac{dx_3}{dt} &= x_3.\end{aligned}$$

Part B (Laplace Transforms and Partial Differential Equations) $[5 \times 5 = 25]$

6. Show that the laplace transform of $\sin \sqrt{t}$ is $\frac{\sqrt{\pi}}{2s^{3/2}} e^{-\frac{1}{4s}}$. [5M]
7. Find the laplace transform of periodic function $f(t)$ with period T and hence find laplace transform of $f(t) = \sin wt$ when $0 < t < \frac{\pi}{w}$, $f(t) = 0$ when $\frac{\pi}{w} < t < \frac{2\pi}{w}$ [5M]
8. Find inverse Laplace transform of $\frac{1}{(s+1)^2(s^2+4)}$. [5M]
9. Solve the integro-differential equation $\frac{dy}{dt} + 4y + 13 \int_0^t y dt = 3e^{-2t} \sin 3t$, $y(0) = 3$ using Laplace transform. [5M]
10. Solve the PDE $u_x = 2u_t + u$, $u(x, 0) = 6e^{-3x}$ by the method of separation of variables. [5M]

Part C (Vector Calculus) $[4 \times 5 + 2 \times 10 = 40]$

11. Using polar integration, find the area of the region R in the xy -plane enclosed by the circle $x^2 + y^2 = 4$, above the line $y = 1$, and below the line $y = \sqrt{3}x$. [5M]
12. Find the center of mass of a solid of constant density δ bounded below by the disk $R : x^2 + y^2 \leq 4$ in the plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$. [5M]
13. A solid of constant density $\delta = 1$ occupies the region of "ice cream cone" D cut from the solid sphere $\delta \leq 1$ by the cone $\phi = \frac{\pi}{3}$. Find the solid's moment of inertia about the z -axis. [5M]
14. Find the line integral $f(x, y, z) = 2xy + \sqrt{z}$ over the helix $r(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$, $0 \leq t \leq \pi$. [5M]
15. Show that $ydx + xdy + 4dz$ is exact and evaluate the integral $\int_{(1,1,1)}^{(2,3,-1)} ydx + xdy + 4dz$ over any path from $(1, 1, 1)$ to $(2, 3, -1)$. [5M]
16. Calculate the outward flux of the vector field $F(x, y) = 2e^{xy} \mathbf{i} + y^3 \mathbf{j}$ across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$. [5M]
17. Find the surface area cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$. [5M]
18. State Stoke's theorem verify for the hemisphere $S : x^2 + y^2 + z^2 = 9, z \geq 0$ its bounding circle $C : x^2 + y^2 = 9, z = 0$ and the field $F = y \mathbf{i} - x \mathbf{j}$. [5M]
19. (a) Calculate the flux of the vector field $F = x^2 \mathbf{i} + 4xyz \mathbf{j} + ze^x \mathbf{k}$ out of the box-shaped region $D : 0 \leq x \leq 3, 0 \leq y \leq 2, 0 \leq z \leq 1$.
(b) Integrate $\text{div} F$ over this region and show that the result is the same value as in part (a), as asserted by the Divergence theorem. [5M]
20. Let W be solid bounded by the cone $z = 2\sqrt{x^2 + y^2}$ and the plane $z = 4$. Assume that the density of the material inside W varies as $\delta(x, y, z) = 5 - z$. Let us calculate the moment of inertia I_z about the z -axis. [5M]

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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, May-2025

Section: B

Course Name: Professional Communication

Date: 01/05/2025

Duration: 3 Hours

Course Code: HU150

Time: 10:00 am-01:00 pm

Max. Marks: 100

ANSWER ALL QUESTIONS

Section A: Descriptive Questions

1. Discuss the statement: "Non-verbal communication speaks louder than words in professional settings." Write an argumentative essay supporting or opposing this view. (7)
2. Prepare a detailed outline for a technical presentation on "The Role of Communication Networks in Enhancing Collaboration." Include a brief introduction, key points, suggested visuals, and the conclusion you aim to achieve. (8)
3. Imagine you are the CEO of a company facing negative rumors circulating via the grapevine. You want to use the grapevine positively to motivate employees. Write:
 - a) A memo you would share informally to counter rumors. (7)
 - b) An agenda notice to establish a healthy grapevine culture in your organization. (7)
4. Select a character from a famous novel, movie, or TV series known for their communication style (e.g., Sherlock Holmes, Don Draper, Leslie Knope and Robin Hood).
 - a) Analyze their verbal and non-verbal communication traits. (6)
 - b) Suggest how they could improve their communication using the 7Cs or other principles from the syllabus. (5)
5. Reflect on a recent group discussion or debate you participated in.
 - a) Analyze your communication strengths and weaknesses during the session. (6)
 - b) Propose a plan to improve your professional communication skills using insights from this syllabus. (5)
6. Think of a situation where poor listening caused a misunderstanding in your personal or professional life.
 - a) Describe the incident and the factors that contributed to ineffective listening. (5)
 - b) Reflect on how employing the principles of good listening could have altered the outcome. (5)
7. Design a game that teaches employees the principles of effective communication.
 - a) Outline the game's concept, rules, and objectives. (7)
 - b) Explain how it will reinforce key concepts like the 7Cs, feedback, and noise reduction. (5)
8. A company's feedback system faced issues: employees feared giving honest feedback, feedback was unclear, and some were ignored, causing frustration. (6)
 1. Tasks: Identify sender- and receiver-oriented barriers, suggest three ways to improve feedback, and explain how confirmatory and corrective feedback can help.

9. Section B: Match the Following

(8)

Column A	Column B
Downward Communication	Exchange among employees of the same hierarchical level
Upward Communication	Communication following official hierarchy
Horizontal Communication	Use of media for large and impersonal audience
Diagonal Communication	Sharing of information among individuals in proximity
Formal Communication	Self Talk
Mass Communication	Reports sent to superiors
Intrapersonal Communication	Information or orders sent to subordinates
Interpersonal Communication	Cross-functional communication across different departments

10. Section C: Fill in the Blanks:

(5)

- The _____ format of business letters aligns all elements to the left margin.
- _____ letters are written to inquire about product details, such as quality, price, and delivery terms.
- A _____ is a formal written communication used to share internal information within an organization.
- The _____ line in a business letter gives the reader a brief idea about the letter's purpose.
- The _____ block format of a business letter positions the complimentary close and signature block to the right.

11. Section D: True or False:

(5)

- A notice usually requires only one copy to be posted.
- Minutes of a meeting can include personal remarks of attendees.
- An agenda must be clear and specific for all authorized participants.
- Circulars are more likely to have legal standing than notices.
- Notices are usually printed in larger typefaces than circulars for readability.

12. Section E: Sentence Improvement:

(3)

- Rewrite this sentence to make it more concise:
"It is hereby informed that the meeting has been scheduled to take place on 28th April, 2025."
- Improve the clarity of this notice sentence:
"Employees of the accounts department will all meet together at the main conference room at 3:00 PM."
- Transform the following into the passive voice:
"The manager will distribute the agenda at the start of the meeting."



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa - 403703

Programme Name: B.Tech.

End Semester Examinations, March-2025

Course Name: **Basics of Mechanical and Civil Engineering**

Course Code: **ME150**

Date: 29-04-2025

Time: 10.00 AM to 1:00 PM

Duration: 180 Minutes

Max. Marks: 100

Instructions: Answer all the questions. Should you think that any data is needed, make an assumption and incorporate it into the solutions. Do not forget to write the appropriate units while answering the questions.

1.

(a) Following gears are in store:

Gear 1:	Gear 2:	Gear 3:	Gear 4:	Gear 5:
Spur gear,	Helical gear,	Spur gear,	Bevel gear,	Spur gear,
Material: Steel	Material: Steel,	Material: Steel	Material: Steel	Material: Steel
Module: 2.5	Module: 2.5	Circular	Module: 2.5	Module: 2.5
Number of	Number of	pitch = 9.424	Number of	Number of
teeth: 24	teeth: 36	Number of	teeth: 36	teeth: 36
		teeth: 36		

The other specifications are assumed to be compatible if they can be mesh. Think about meshing the input gear (gear 1) with every other piece of gear in the store. Are they meshable? If so, describe the result. If meshing was not possible, then explain why. **(4 M)**

(b) A train of spur gears is required to give a total reduction of 250 to 1 in four steps. No pinion is to have less than 20 teeth and the diametral pitches are to be 5 for the first step, 3.5 for the second, 2.5 for the third and 1.5 for the fourth. The center distances must not involve recurring decimals. Find the numbers of teeth, the pitch-circle diameters and the center distances for a suitable train of gears. **(5 M)**

(c) Derive the expression of length for an open-belt drive. Considering the tension induced due to centrifugal inertial force as $T_c = mv^2$ (where m : mass of the belt per unit length and v is the velocity of the belt), deduce the condition for maximum power transmission. **(5 M)**

(d) Determine the belt length of an open belt to connect the 5 cm and 10 cm diameter pulley at a centre distance of 75 cm. **(5 M)**

2.

(a) Somebody proposes the following system to cool a house in the summer: Compress the regular outdoor air, let it cool back to the outdoor temperature, pass it through a turbine, and discharge the cold air leaving the turbine into the house. From a thermodynamic point of view, is the proposed system sound? **(3 M)**

- (b) A reversible heat engine operates between two reservoirs at temperatures of 600°C and 40°C . The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 40°C and -20°C . The heat transfer to the heat engine is 2000 kJ and the net-work output for the combined engine refrigerator is 360 kJ . (i) Calculate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C . (ii) Reconsider (i) given that the efficiency of the heat engine and the C.O.P. of the refrigerator are each 40 per cent of their maximum possible values. (7 M)

3.

- (a) A rectangular block with mass 2 kg is sliding down an inclined plane at angle 30° . The block is separated from the plane by a lubrication oil with dynamic viscosity 0.25 Ns/m^2 with spacing of 1 mm . The base of the block has area $20\text{ cm} \times 50\text{ cm}$. Determine the terminal velocity (when acceleration is zero) of the block as it slides down the plane in Fig. 3(a).

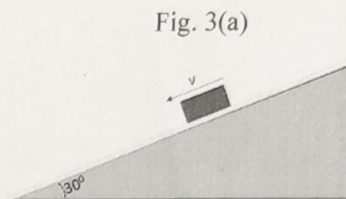


Fig. 3(a)

(5 M)

- (b) (a) The roof of an electrically heated home is 6 m long, 8 m wide, and 0.25 m thick, and is made of a flat layer of concrete whose thermal conductivity is $k=50.8\text{ W/m}\cdot\text{K}$. The temperatures of the inner and the outer surfaces of the roof one night are measured to be 15°C and 4°C , respectively, for a period of 10 hours . Determine (a) the rate of heat loss through the roof that night and (b) the cost of that heat loss to the home owner if the cost of electricity is $\text{₹}8/\text{kWh}$. (6 M)

- (c) Consider steady heat conduction through a large plane wall of thickness x_1 , x_2 and x_3 and area A as shown in the Fig.3(d). The temperature profile across the across thickness is given in the Fig. Based on the temperature profile given can you compare the thermal conductivity of the materials in the wall stack. (2 M)

4.

- (d) If a highly viscous liquid is placed on the hot steel surface. Explain about whether the fictitious experiment shown in Fig. 3(e) qualifies as convection or conduction. (2 M)

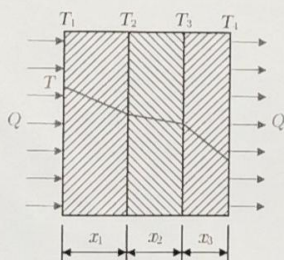


Fig. 3(d)

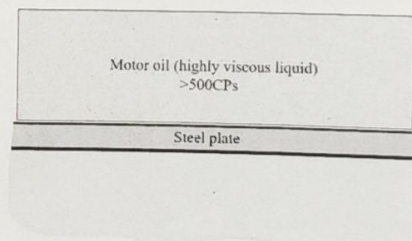


Fig. 3(e)

4.

- (a) The minimum pressure and temperature in an Otto cycle are 100 kPa and 27°C . The amount of heat added to the air per cycle is 1500 kJ/kg . (i) Determine the pressure and temperature at all points of the Otto cycle. (ii) Also calculate the specific work and thermal efficiency of the cycle for a compression ratio of $8:1$. Take for air $C_v=0.72\text{ kJ/kg}\cdot\text{K}$ and $\gamma=1.4$. (5 M)

- (b) Find the expression of Air-Standard efficiency of Diesel Combustion Cycle. Compare the efficiency of the efficiency of diesel cycle with that of otto cycle. What can we anticipate if the otto cycle compression ratio is raised to match the diesel engine? By mistake, someone puts gasoline into a diesel engine. What can we anticipate? **(10M)**

5.

- (a) A 400 mm long bar has rectangular cross-section $10 \text{ mm} \times 30 \text{ mm}$. This bar is subjected to (i) 15 kN tensile force on $10 \text{ mm} \times 30 \text{ mm}$ faces, and (ii) 80 kN compressive force on $10 \text{ mm} \times 400 \text{ mm}$ faces, and (iii) 180 kN tensile force on $30 \text{ mm} \times 400 \text{ mm}$ faces. Find the change in volume if $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$. **(5M)**

- (b) Steel bar 2.5 m long has a circular cross section of diameter $d_1 = 0.02 \text{ m}$ over one-half of its length and diameter $d_2 = 0.0125 \text{ m}$ over the other half. The modulus of elasticity $E = 200 \text{ GPa}$. (a) How much will the bar elongate under a tensile load $P = 20 \text{ kN}$? (b) If the same volume of material is made into a bar of constant diameter d and length 2.5 m, what will be the elongation under the same load P ? **(10M)**

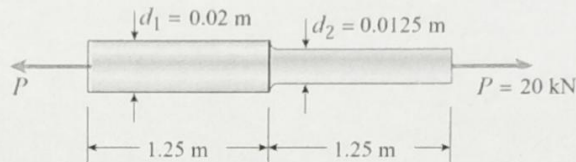


Fig. 5(b)

- (c) Define the following (a) casting process and pattern, (b) depth is of cut and feed in machining, (c) solid state welding, TIG and MIG welding? **(6M)**

6.

- (a) Define built-up area. **(2M)**
- (b) List out the
- (i) types of building Plans. **(2M)**
 - (ii) processes for setting out of building. **(2M)**
 - (iii) types of flooring and roofing. **(2M)**
 - (iv) types of plaster and finishing. **(2M)**
 - (v) types of conventional and modern building materials. **(2M)**
 - (vi) essential elements of hydropower station. **(1M)**
 - (vii) classification of head for hydropower plants **(1M)**
 - (viii) sources of raw water. **(1M)**
 - (ix) elements of water supply system. **(2M)**
 - (x) components of roof top rainwater harvesting system. **(2M)**
 - (xi) methods of recharging ground water aquifers. **(1M)**



NATIONAL INSTITUTE OF TECHNOLOGY, GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, May 2025

Roll no	2	4	C	S	E	1	0	2	7
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Course Name: Engineering Chemistry (Regular)

Date: 02.05.2025

Duration: 3 Hours

Course Code: CY150

Time: 10 AM

Max. Marks: 100

1. Answer all the questions.
2. Answers for sub-questions have to be labelled properly
3. All calculations/workouts must be shown in the answer script properly

1 Write the synthesis of the following polymers. 6 Marks

- a) Polyvinyl chloride vinyl acetate co-polymer b) Poly amide co-polymer, c) poly ester co-polymer

2 Write the synthesis of the following polymers 6 Marks

- a) Polybutadiene, b) Gutta-percha

Based on their structures, comment on which polymer is expected to have a higher T_g value and why

3 Compare the mechanism of synthesis of polypropylene and polyacrylonitrile. Write a most suitable mechanism of synthesis of both stepwise. 6 Marks

4 Calculate the molecular mass of the given polymer from the data given below by using two known methods. Also, find out the polydispersity index (PDI) for the given polymer. 7 Marks

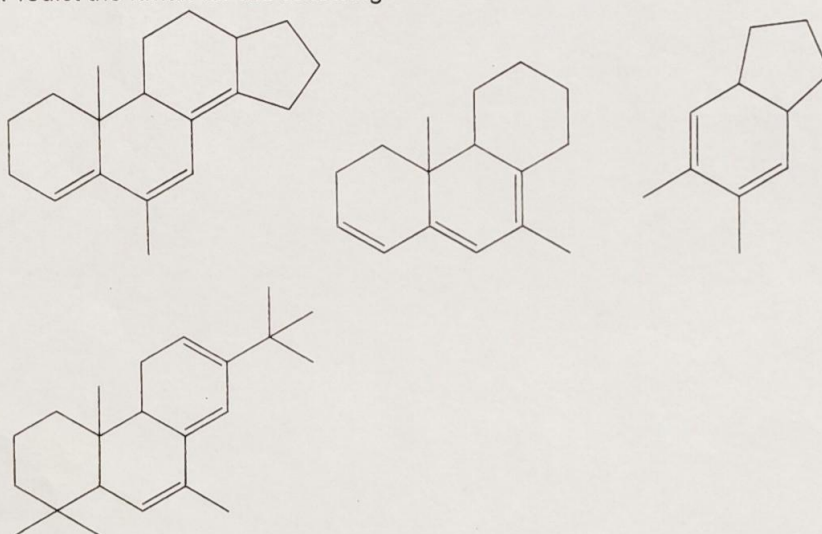
Number	Molecular Weight (g/mol)
500	10,000
10	250,000
3	100,000,000
6	15,000
120	3,000

5 The following is the data obtained by the titration of the mixture of A and B with C. Calculate the % A, B, and C, tabulate them, and mark these points on the phase diagram. Draw a binodal curve and mark the plait point on the phase diagram. 8 Marks

A (mL)	B (mL)	C (mL)
1	2	7
3	4	3

6	3	1
5	2	3
4	4	2
2	2	6

- 6 Draw, calculate degrees of freedom, and explain the phase diagram of a system involving metal extraction from the ore 5 Marks
- 7 Draw, calculate degrees of freedom, and explain the phase diagram of a system involving allotropes, and comment and prove why 4 phases cannot exist together 6 Marks
- 8 Draw the phase diagrams of two systems, one involving a non-decomposition melting point and the other with a decomposition point. Mark, name, and label all the important points. Find out the degrees of freedom at these two points. (Note: No need to calculate degrees of freedom at other points, curves, and areas.) 6 Marks
- 9 Predict the λ_{max} for the following molecules 8 Marks



- 10 Calculate the number of normal vibrational modes possible for SO_2 and HCN molecules. Identify each of them and predict whether the motion is IR active or not. Justify your answer. 6 Marks
- 11 Calculate the absorption frequency corresponding to the $\text{—C}\equiv\text{N}$ stretching vibration treating the group as a simple diatomic C—N molecule with a force constant of $k = 1.5 \times 10^3 \text{ N/m}$ 4 Marks
- 12 A student recorded the UV-Vis spectrum of p-nitroaniline in benzene as the solvent. The student noticed a complex spectrum. He went to his professor and asked for his advice. This professor advised him not to use benzene as a solvent to record UV-Vis spectrum and recommended him to use a different solvent. 3 Marks
- Explain why benzene should not be used to record UV-Vis spectrum
 - Suggest two solvents that can be used to record UV-Vis spectrum
- 13 Predict low and high-resolution NMR for the following molecules mentioning appropriate relative signal intensities. 12 Marks

1. Isopropyl Bromide
2. Ethyl bromide
3. 1-Bromo 2-Iodoethane
4. Methyl acetate

- 14 Describe corrosion due to carbon dioxide and Acid with detailed chemical reactions. Discuss the source and methods to prohibit corrosion as well. 7 Marks
- 15 Discuss the caustic embrittlement with a detailed chemical reaction. 5 Marks
- 16 Describe the cold soda lime process of water treatment. 5 Marks

****ALL THE BEST****



National Institute of Technology Goa

Programme Name: B.Tech.

End Semester Examinations, April, 2025

Course Name: **Basics of Electronics Engineering**

Date: 28/04/2025

Duration: 3 Hour

Course Code: **EC150**

Time: 10 AM - 1 PM

Max. Marks: 100

ANSWER ALL QUESTIONS

Instructions

- Answer all the questions. Assume suitable values if data is insufficient

Part A: 4 Marks Questions

1. Why are NAND and NOR gates called universal gates? Justify your answer with the help of examples.
2. Convert $C3DF$ to binary. What is the 2's complement to the result.
3. What are the basic building blocks of communication system.
4. What are input and output characteristics of depletion type MOSFET. Write down current and voltage relationship.
5. What is the difference between message signal and carrier signal?
6. Plot input and output characteristic of BJT Common-^{Base} ~~Emitter~~ (CB) configuration?
7. What is clamper circuit?

Part B: 8 Marks Questions

1. $X = 10101110$, $Y = 10011101$. Perform the subtraction (a) $X - Y$ (b) $Y - X$ using 2's complement and 1's complement.
2. Design a suitable amplifier circuit using an operational amplifier (op-amp) with a gain of 5. A $3\text{mV}_{\text{p-p}}$ (1kHz) signal is applied to the circuit. Draw the input and output waveforms.
3. Simplify the Boolean function using K-map: $F(A, B, C, D) = \sum(1, 3, 4, 7, 8, 11, 15)$
4. Find out output voltage in Figure. 1

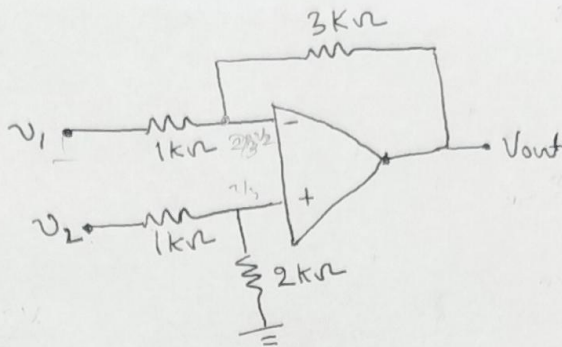


Figure 1

5. In a JFET $I_{DSS} = 10\text{mA}$ and Pinch-off voltage $V_P = -5\text{V}$. Calculate the Drain current for $V_{GS} = 0, -2, -4\text{V}$.
6. What is the only input combination that: (a) Will produce a logic '1' at the output of an eight-input AND gate? (b) Will produce a logic '0' at the output of a four-input NAND gate? (c) Will produce a logic '1' at the output of an eight-input NOR gate? (d) Will produce a logic '0' at the output of a four-input OR gate?

Part C: 12 Marks Questions

1. What is a modulation? Why it is important? Discuss about Analog and Digital Modulation Techniques.
2. (a) For the following network Figure. 2, $I_B = 0.01\text{mA}$. Calculate I_C and V_{CE}

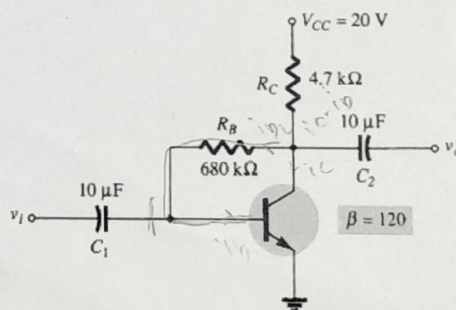


Figure 2

- (b) Draw the truth table of the logic circuit shown below Figure. 3

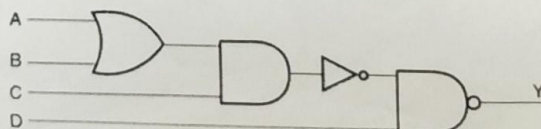


Figure 3

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National Institute of Technology Goa

Cuncolim, Salcete, South Goa - 403703

Program Name: B.Tech., Year: First

End-Semester Examinations, April-May 2025

Course Name: **Health and Happiness**

Date: 30/04/2025

Duration: 180 Minutes

Course Code: **HU151**

Time: 10.00 AM - 01.00 PM

Max.s: 100

Answer all questions. Please ensure that your answers are written neatly and organized in a point-wise format for clarity and ease of understanding.

1. Answer the following multiple choice questions.

[1 × 30 = 30]

1. Which of the following is NOT a quality attribute of food?

- (A) Appearance (B) Packaging (C) Brand ambassador (D) Texture

2. From which French word is the term "sautéing" derived?

- (A) To Burn (B) To Soften (C) To Jump (D) To Mix

3. What is meant by food adulteration?

- (A) Adding nutrients to food (C) Intentionally degrading food quality
(B) Enhancing taste using spices (D) Cooking food at high temperatures

4. What is the ideal temperature range for poaching?

- (A) 100°C (B) 90–95°C (C) 60–80°C (D) 30–40°C

5. Golden Rice is enriched with which vitamin?

- (A) Vitamin B12 (B) Vitamin A (C) Vitamin D (D) Vitamin C

6. Which asana is common to both Surya Namaskar and Chandra Namaskar?

- (A) Tadasana (C) Padahastasana
(B) Ardha Chandrasana (D) Shavasana

7. Which path of Yoga is referred to as the Path of Devotion?

- (A) Karma Yoga (B) Bhakti Yoga (C) Jnana Yoga (D) Raja Yoga

8. What does Pratyahara refer to?
- (A) Breathing control (C) Sensory withdrawal
(B) Body postures (D) Enlightenment
9. Which of the following is a benefit of practicing Ashtanga Yoga?
- (A) Weakens the spine (C) Improves blood circulation
(B) Reduces metabolism (D) Causes fatigue
10. Which nutrient is essential for building muscles and tissues?
- (A) Sugar (B) Protein (C) Water (D) Fiber
11. What aspect of well-being does "Pranayama" primarily deal with?
- (A) Meditation (B) Breathing control (C) Physical posture (D) Fasting
12. Which phytochemical is abundantly present in carrots?
- (A) Isoflavones (B) Beta-carotene (C) Phytosterol (D) Anthocyanidins
13. Which of the following best describes the primary function of food?
- (A) Entertainment (C) Social media content
(B) Providing nutrients and energy (D) None of the above
14. What is the literal meaning of the word "Yoga"?
- (A) Balance (B) To join or unite (C) Exercise (D) Meditation
15. BMI is calculated using which two parameters?
- (A) Weight and age (C) Calories and fat
(B) Weight and height (D) Carbohydrates and protein
16. Who is known for popularizing modern Ashtanga Yoga?
- (A) B.K.S. Iyengar (B) Ramdev Baba (C) K. Pattabhi Jois (D) Sadhguru
17. A balanced diet consists of which combination of nutrients?
- (A) Only carbohydrates (C) Fast food items
(B) Protein, fat, fiber, and carbohydrates (D) None of the above

18. Why is food adulteration typically practiced?

- (A) To increase food safety
- (B) To reduce flavor
- (C) To increase profit and reduce cost
- (D) To improve nutrition

19. Which Yama in Yoga philosophy stands for “non-violence”?

- (A) Satya
- (B) Ahimsa
- (C) Aparigraha
- (D) Brahmacharya

20. What is one important function of water in the human body?

- (A) Causing disease
- (B) Detoxifying the body
- (C) Increasing sugar levels
- (D) Reducing calcium levels

21. What is the primary focus of outcome evaluation?

- (A) Food packaging only
- (B) Overall process implementation
- (C) Specific results or objectives
- (D) External appearance of food

22. Which cooking method is best suited for dietary food using minimal oil?

- (A) Grilling
- (B) Sautéing
- (C) Boiling
- (D) Deep frying

23. Which attribute of food includes its odor and taste?

- (A) Texture
- (B) Flavor
- (C) Nutrition
- (D) Packaging

24. Lack of good nutrition can result in which of the following?

- (A) Good health
- (B) Malnutrition and diseases
- (C) Only weight loss
- (D) Improved mental sharpness

25. Which of the following is considered a macronutrient?

- (A) Iron
- (B) Vitamin C
- (C) Protein
- (D) Iodine

26. Which is a recognized method for evaluating food quality?

- (A) Taste test only
- (B) Guesswork
- (C) Formative evaluation
- (D) Spoilage rate

27. In genetically modified tomatoes, removal of which gene prevents softening?

- (A) Polygalacturonase
- (B) Lycopene
- (C) Flavonoid
- (D) Amylopectin

28. Which of the following is NOT classified as a limb of Ashtanga Yoga?

- (A) Yama (B) Niyama (C) Tadasana (D) Samadhi

29. How is nutrition best defined?

- (A) Cooking style (C) Study of food and its impact on the body
(B) Absorption of nutrients (D) A type of food

30. Anthocyanidins are commonly found in which type of food?

- (A) Red and purple berries (C) Cucumbers
(B) Bananas (D) Onions

2. Answer the following in brief.

[2 × 8 = 16]

1. Why is water essential for the human body?
2. Name any two commonly found food adulterants.
3. What is Ashtanga Yoga?
4. Differentiate between Dhyana and Dharana.
5. How do phytochemicals support the immune system?
6. What is meant by genetically modified (GM) food?
7. What is the purpose of HACCP in food safety?
8. Mention any two risks associated with genetically modified food.

3. Write short notes on the following.

[5 × 5 = 25]

1. Codes of good conduct for Sadvritta
2. Cooking methods
3. Effect of fast food or junk food on our body
4. Quality attributes of food and the factors affecting it
5. What is BMI? Calculate BMI for the following data: Height: 1.95 m, Weight: 80 kg.

4. Answer the following in sufficient details.

[8 × 3 = 24]

1. Define Health. What are the signs of a healthy lifestyle? Explain the concept of Circadian Rhythm. Discuss the relationship between Circadian Rhythm and digestion. [2 + 2 + 2 + 2 = 8]
2. What is Food Adulteration? Explain the different types of food adulteration. Discuss the reasons why food adulteration is practiced. [2 + 3 + 3 = 8]
3. How many Ritus (seasons) are described in traditional practices? Name the Ritus. What is the recommended diet during Sharad Ritu? What lifestyle practices are suggested for Sharad Ritu? [2 + 3 + 3 = 8]

5. Explain Tratak. What are the advantages of Tratak.

[5]